

AG

Avinash

Group App

A
comprehensive
ERPNext
customization
providing
automated tax
calculations
PLATFORM
(Excise Duty,
ERPNext),
BRANCH
VAN APPS),
develop
customer
DATE
credit limit
DOCTYPES
enforcement,
MODIFIED
multi-step
approval
workflows, and
specialized
financial
reports.

Table of Contents

1. Overview & Purpose

2. Features at a Glance (Non-Technical)

2.1 Tax Auto-Calculation 2.2 Customer Credit Limits 2.3 Approval Workflows 2.4 Custom Reports

3. Technical Architecture

3.1 App Structure 3.2 Hooks Configuration

4. All Custom Fields — Complete Reference

4.1 Customer 4.2 Purchase Invoice 4.3 Purchase Order 4.4 Purchase Receipt 4.5 Supplier Quotation 4.6 Sales Invoice 4.7 Other

5. Tax Calculation Logic

6. Credit Limit Validation

7. Workflow & Admin Bypass

8. Custom Reports

9. JavaScript Client-Side Logic

Overview & Purpose

The **Avinash Group App** is a custom ERPNext module that extends the standard Frappe/ERPNext platform with business-specific functionality including tax compliance (Excise Duty, VAT, TDS), customer credit management, procurement approvals, and financial reporting.

Tax Automation

Automatic calculation of Excise Duty, VAT, and TDS on purchase and sales documents with real-time updates.

Credit Control

Enforces customer credit limits by bill count, outstanding days, and total exposure amount.

Approval Workflows

Multi-step approval for Purchase Orders and Material Requests with admin bypass capability.

Custom Reports

Vehicle expense tracking, supplier price comparison, and extended P&L hierarchy reporting.

Scope of Customization

The app modifies **13 ERPNext doctypes** through custom fields, hooks, and overrides:

- **Purchase:** Purchase Invoice, Purchase Order, Purchase Receipt, Supplier Quotation (+ Item child tables)
- **Sales:** Sales Invoice (+ Item child table)
- **Master Data:** Customer
- **Other:** Packed Item, Purchase Taxes and Charges

Features at a Glance

This section is written for non-technical readers — no programming knowledge required.

2.1 Tax Auto-Calculation

When you create or edit a purchase or sales document, the system automatically computes taxes for each line item and shows running totals at the bottom. No manual tax calculations needed.

1 Enter item & net amount

Fill in item, quantity, and rate as normal in ERPNext.



2 Enter Excise Value (if applicable)

Type the excise duty amount. System adds it to net amount → "Total incl. Excise".



3 VAT calculated automatically

System applies the VAT rate from item's tax template. Switch to "Amount" mode for manual entry.



4 TDS deducted (purchases only)

If Tax Withholding Category is set and "Apply TDS" is ticked on the line, TDS is auto-deducted.



5 Document totals updated

Bottom of document shows: Total Excise, Total VAT, Total TDS, Grand Total — all automatic.

NOTE ON SALES INVOICE

TDS is **not** applicable on Sales Invoices. Only Excise Duty and VAT are calculated there.

2.2 Customer Credit Limits

The system checks three credit limits when a Sales Invoice is saved. Configured on the Customer record.

LIMIT TYPE	WHAT IT CHECKS	FIELD ON CUSTOMER
Bill Count Limit	Maximum number of unpaid invoices allowed	Max Unpaid Bills
Days Limit	Maximum days any invoice can remain unpaid	Max Unpaid Days
Amount Limit	Maximum total outstanding amount	Max Credit Amount

WHEN LIMIT IS EXCEEDED

The system blocks saving the Sales Invoice with a clear error. Advance payments already received are automatically deducted before checking limits.

2.3 Approval Workflows

- **Purchase Order** — Requires a designated PO Reviewer to approve before submission.
- **Material Request** — Goes through a One-Line Approver workflow.

ADMIN BYPASS

Users with the **Administrator** role can bypass workflow conditions and move documents directly to any state — useful for corrections and urgent situations.

2.4 Custom Reports

REPORT NAME	WHAT IT SHOWS	WHO USES IT
Supplier Quotation Comparison	Prices from all suppliers per item, side-by-side	Procurement Team
Vehicle Expense Report	Fuel, repairs, and other costs per vehicle	Fleet / Finance
Profit & Loss Hierarchy	Extended P&L with growth %, margin %, account-level filter	Management / Finance

Technical Architecture

3.1 App Structure

```

avinashgroup_app/
├── avinash_group_app/
│   ├── custom/                                # Custom field JSON (13 files)
│   │   ├── customer.json
│   │   ├── purchase_invoice.json / purchase_invoice_item.json
│   │   ├── purchase_order.json / purchase_order_item.json
│   │   ├── purchase_receipt.json / purchase_receipt_item.json
│   │   ├── supplier_quotation.json / supplier_quotation_item.json
│   │   ├── sales_invoice.json / sales_invoice_item.json
│   │   ├── packed_item.json
│   │   └── purchase_taxes_and_charges.json
│   └── report/
│       ├── profit_and_loss_hierarchy/
│       ├── avinas_vehicle_expense/
│       └── custom_supplier_quotation_comparison/
├── custom_code/
│   ├── common/purchase_taxes_handler.py      # Core tax engine
│   ├── SalesInvoice/salesinvoice_taxes.py   # Sales tax + credit limit
│   ├── Override/material_request.py          # Workflow bypass
│   ├── Override/purchase_order.py            # Workflow bypass
│   └── workflow_admin_bypass.py              # Admin bypass logic
└── public/js/
    ├── purchase_taxes_common.js purchase_invoice.js
    ├── sales_invoice.js             purchase_order.js
    ├── material_request.js          journal_entry.js
    └── global_filter.js

```

3.2 Hooks Configuration

DOCTYPE	EVENT	HANDLER
Purchase Invoice	before_save / validate / before_submit	purchase_taxes_handler / excise_ledger
Purchase Order	before_save / validate	purchase_taxes_handler
Purchase Receipt	before_save / validate	purchase_taxes_handler
Supplier Quotation	before_save / validate	purchase_taxes_handler
Sales Invoice	before_save / validate	salesinvoice_taxes

Overridden Doctype Classes

DOCTYPE	OVERRIDE CLASS	PURPOSE
Material Request	custom_code.Override.material_request.MaterialRequest	Workflow admin bypass
Purchase Order	custom_code.Override.purchase_order.PurchaseOrder	Workflow admin bypass

Whitelisted Method Overrides

METHOD	OVERRIDE
get_transitions	custom_code.workflow_admin_bypass.get_transitions
apply_workflow	custom_code.workflow_admin_bypass.apply_workflow
RFQ create_supplier_quotation	templates.pages.rfq.create_supplier_quotation

All Custom Fields — Complete Reference

4.1 Customer

FIELD NAME	LABEL	TYPE	PURPOSE
custom_company	Company	Link	Links customer to a company. Required. Global search enabled.
custom_abbr	Company Abbr	Data	Auto-fetched abbreviation from linked company.
custom_bill_count	Max Unpaid Bills	Int	Max number of unpaid invoices allowed for this customer.
custom_amount_limit	Max Credit Amount	Float	Max total outstanding amount allowed in currency.
custom_days_limit	Max Unpaid Days	Int	Max days any invoice can remain unpaid.
custom_created_by	Created By	Link(User)	Audit — auto-set, read-only.
custom_created_on	Created On	Datetime	Audit — auto-set, read-only.
custom_modified_by	Modified By	Link(User)	Audit — auto-set, read-only.

4.2 Purchase Invoice

Item (child table)

FIELD NAME	LABEL	TYPE	PURPOSE
custom_subtype	Vehicle	Link(Vehicle)	Assign this line item expense to a specific vehicle.
custom_excise_value	Excise Value	Currency	Manual excise duty for this item. No copy on amendment.
custom_total	Total (incl. Excise)	Currency	net_amount + excise_value. Calculated, read-only.
custom_vat_apply_on	VAT Apply On	Select	"Percentage (%)" or "Amount". Default: Percentage.
custom_vat_rate	VAT Rate (%)	Percent	VAT rate. Read-only in % mode (auto-fetched).
custom_vat_amount	VAT Amount	Currency	Calculated VAT. Manual if "Amount" mode.

FIELD NAME	LABEL	TYPE	PURPOSE
custom_tds_apply_on	TDS Apply On	Select	"Percentage (%)" or "Amount". Default: Percentage.
custom_tds_rate	TDS Rate (%)	Percent	Fetches from Tax Withholding Category.
custom_tds_amount	TDS Amount	Currency	Calculated TDS. Active only when apply_tds = 1.

Header fields

FIELD NAME	LABEL	PURPOSE
custom_abbr	Company Abbr	Auto-fetched company abbreviation.
custom_total_amount_including_excise	Total Incl. Excise	Sum of all item custom_total values.
custom_total_excise_amount	Total Excise	Sum of all item excise values.
custom_total_vat_amount	Total VAT	Sum of all item VAT amounts.
custom_total_tds_amount	Total TDS	Sum of TDS where apply_tds = 1.
custom_created_by / custom_created_on / custom_modified_by	Audit Fields	Read-only audit trail.

4.3 Purchase Order

Item fields

Identical to Purchase Invoice Item: custom_excise_value , custom_total , custom_vat_apply_on , custom_vat_rate , custom_vat_amount , custom_tds_apply_on , custom_tds_rate , custom_tds_amount .

Header fields

FIELD NAME	LABEL	TYPE	PURPOSE
custom_abbr	Company Abbr	Data	Auto-fetched.
custom_approver	PO Reviewer	Link(User)	Required approver for this Purchase Order.
custom_po_request_approver	Approver List	Table	Multi-approver workflow table.
custom_remarks	Remarks	Small Text	Additional notes for this PO.

FIELD NAME	LABEL	TYPE	PURPOSE
custom_total_amount_including_excise	Total Incl. Excise	Currency	Aggregated total with excise.
custom_total_excise_amount	Total Excise	Currency	List view.
custom_total_vat_amount	Total VAT	Currency	List view.
custom_total_tds_amount	Total TDS	Currency	List view.

4.4 Purchase Receipt

Item fields: Same as Purchase Invoice Item (excise, VAT, TDS fields).

Header fields: Same as Purchase Invoice header (abbr, totals, audit fields).

4.5 Supplier Quotation

Item fields

Same tax fields as other purchase item tables (excise, VAT, TDS).

Header fields

FIELD NAME	LABEL	TYPE	PURPOSE
custom_preferred_quotation	Preferred Quotation	Check	Mark as preferred quote. Editable after submit. Used in comparison report.
custom_total_amount_including_excise	Total Incl. Excise	Currency	List view.
custom_total_excise_amount	Total Excise	Currency	List view.
custom_total_vat_amount	Total VAT	Currency	List view.
custom_total_tds_amount	Total TDS	Currency	List view.

4.6 Sales Invoice

IMPORTANT

Sales Invoice has **no TDS fields**. Only Excise Duty and VAT apply.

Item fields

FIELD NAME	LABEL	TYPE	PURPOSE
<code>custom_excise_value</code>	Excise Value	Currency	Manual excise duty per item.
<code>custom_total</code>	Total (incl. Excise)	Currency	<code>net_amount + excise_value</code> . Calculated.
<code>custom_vat_apply_on</code>	VAT Apply On	Select	Percentage (%) or Amount mode.
<code>custom_vat_rate</code>	VAT Rate (%)	Percent	VAT rate from item tax template.
<code>custom_vat_amount</code>	VAT Amount	Currency	Calculated VAT.

Header fields

FIELD NAME	LABEL	PURPOSE
<code>custom_total_amount_including_excise</code>	Total Incl. Excise	Aggregated total with excise.
<code>custom_total_excise_amount</code>	Total Excise	Sum of all excise values.
<code>custom_total_vat_amount</code>	Total VAT	Sum of all VAT amounts.
<code>custom_total_amount</code>	Total Amount	Total excluding excise.

4.7 Packed Item & Purchase Taxes and Charges

DOCTYPE	FIELD	CHANGE	PURPOSE
Packed Item	rate	<code>read_only = 1</code>	Prevents manual editing of rate in packed items.
Purchase Taxes and Charges	<code>add_deduct_tax</code>	Default = "Deduct"	TDS rows default to deduct mode automatically.

Tax Calculation Logic

Item-Level Formulas

```

custom_total      = base_net_amount + custom_excise_value

VAT (% mode)      = custom_total x (custom_vat_rate / 100)
VAT (Amount mode) = manually entered value (kept as-is)

TDS (% mode)      = custom_total x (custom_tds_rate / 100)
                  [ ONLY IF: apply_tds = 1 AND tax_category set ]
TDS (Amount mode) = manually entered value

```

Document Totals

```

custom_total_amount_including_excise = SUM( custom_total for all items )
custom_total_excise_amount           = SUM( custom_excise_value for all items )
custom_total_vat_amount              = SUM( custom_vat_amount for all items )
custom_total_tds_amount              = SUM( custom_tds_amount where apply_tds = 1 )

```

GL / Taxes Table Mapping

TAX TYPE	GL ACCOUNT PREFIX	DIRECTION	CONDITION
Excise Duty	348204	Add	If excise > 0
VAT	VAT	Add	If VAT > 0
TDS	From Tax Withholding Category	Deduct	apply_tds = 1 AND tax_category set

Execution Order (before_save)

1

Set defaults

Ensure VAT/TDS apply_on fields have values.



2

calculate_custom_total()

Per item: custom_total = base_net_amount + custom_excise_value



3 calculate_total_amount_including_excise()

Sum all custom_total → header field



4 calculate_item_vat_amounts()

Compute VAT per item based on mode



5 calculate_item_tds_amounts()

Compute TDS per item (with conditions)



6 update_taxes_table()

Sync Excise / VAT / TDS rows in Taxes table for GL posting



7 calculate_taxes_and_totals()

ERPNext standard method recalculates grand total

Credit Limit Validation

Triggered on the `validate` event of **Sales Invoice** via `validate_sales_invoice()` in `salesinvoice_taxes.py`.

Validation Flow

- 1 Fetch all outstanding Sales Invoices**
Gets all submitted, unpaid/partially paid invoices ordered oldest to newest (FIFO).
- 2 Fetch unallocated advance payments**
Payment Entries with unallocated amounts — applied to oldest invoices first.
- 3 Check Bill Count Limit**
Effective outstanding invoices > `custom_bill_count` → Error.
- 4 Check Days Limit**
Any invoice outstanding > `custom_days_limit` days → Error.
- 5 Check Amount Limit**
Total outstanding > `custom_amount_limit` → Error.

ADVANCE PAYMENT TREATMENT

Advances are applied FIFO: oldest unpaid invoice receives advance first, reducing effective outstanding before limit checks run.

Configuration

CUSTOMER FIELD	CONTROLS	IF BLANK
<code>custom_bill_count</code>	Max unpaid invoices allowed	No limit enforced
<code>custom_days_limit</code>	Max days an invoice can be unpaid	No limit enforced
<code>custom_amount_limit</code>	Max total outstanding amount	No limit enforced

Workflow & Admin Bypass

Active Workflows

WORKFLOW NAME	APPLIES TO	ADMIN BYPASS
Purchase Order Workflow	Purchase Order	Yes
Material Request One-Line Approver	Material Request	Yes

Admin Bypass Mechanism

The file `workflow_admin_bypass.py` overrides two core Frappe whitelist functions:

- `get_transitions(doc)` — When user is Administrator, returns all workflow transitions regardless of role or condition checks.
- `apply_workflow(doc, action)` — When user is Administrator, applies the action bypassing all validation constraints.

SECURITY NOTE

Bypass is scoped strictly to the Administrator role. Regular users still go through full workflow with role and condition enforcement.

Purchase Order Approver Setup

Each PO has a `custom_approver` (Link to User) that must be filled. This user is the designated reviewer for that specific PO. The `custom_po_request_approver` table supports multi-approver scenarios.

Custom Reports

8.1 Supplier Quotation Comparison

Pivot-style report showing item prices from multiple suppliers side-by-side. Helps procurement pick the best-priced supplier for each item.

DIMENSION	CONTENT
Rows	One row per item
Columns	One column per supplier
Cell Value	Configurable: base_rate / base_amount / rate / amount
Summary Rows	Net Total, Discount, Taxable Amount, VAT/Tax, Invoice Amount

Filters: Company, RFQ, Item Code, Supplier, Date Range, Include Expired toggle, Price Field selector.

Special: Preferred quotations (custom_preferred_quotation) are highlighted. Whitelist function `set_default_supplier()` lets users set default supplier from the report.

8.2 Vehicle Expense Report

Tracks expenses per vehicle from Purchase Invoices (custom_subtype = vehicle) and Journal Entries.

CATEGORY	GL ACCOUNT CODES
Fuel Allowance	547136, 547137, 547138
Repair & Maintenance	549301, 549302, 549303
Other Vehicle Expenses	542902, 552002

8.3 Profit and Loss Hierarchy

Extended ERPNext P&L report with:

- **Growth View** — Period-over-period percentage change
- **Margin View** — Expense items as % of revenue
- **Account Level Filtering** — Hide group accounts, focus on leaf-level details
- **Period-Based** — Monthly, quarterly, or annual breakdown

JavaScript Client-Side Logic

Client-side JS files provide real-time UI updates, field toggling, and tax recalculation as users type in forms.

FILE	APPLIES TO	KEY FUNCTIONALITY
<code>purchase_taxes_common.js</code>	All purchase doctypes	Core engine: recalculates excise/VAT/TDS on field change, toggles read-only modes, triggers server recalc
<code>purchase_invoice.js</code>	Purchase Invoice	PI-specific UI handlers and field triggers (24KB)
<code>sales_invoice.js</code>	Sales Invoice	VAT/Excise handlers, credit limit UI feedback
<code>purchase_order.js</code>	Purchase Order	Approver field handling, PO-specific triggers
<code>material_request.js</code>	Material Request	Workflow and approver UI handling
<code>journal_entry.js</code>	Journal Entry	Custom vehicle subtype linking
<code>global_filter.js</code>	All forms	Global filtering and company-based context injection

CLIENT VS SERVER LOGIC

- Client-side JS gives **immediate visual feedback** as users fill in fields.
- Server-side Python recalculates on **before_save** for data integrity (works for API imports too).
- Both layers run the same formulas — the server is always the source of truth.